

The Dependency Architecture

All Fifteen Directed Relationships Between the Six Pillars — How They Work, What Breaks When They Are Violated, and Why the Sequence Is Not Optional

Chapter 1A is the analytical companion to Chapter 1. Where Chapter 1 defines the six pillars and introduces the three types of inter-pillar relationship, this chapter develops all fifteen directed dependency relationships in full — their mechanisms, their failure modes, and the Vermont evidence that tests each relationship against real-world experience. Chapter 2 draws on this dependency architecture to derive the execution sequence. This chapter is designed as a reference: readers can work through it sequentially or navigate directly to the dependency relationships most relevant to their immediate decisions.

I. How to Read This Chapter

Each of the fifteen dependency relationships is documented using a consistent structure: the upstream pillar and downstream pillar involved; the type of relationship (enables, requires, or constrains — as defined in Chapter 1, Section V); the mechanism by which the upstream pillar affects the downstream pillar's function; the failure mode produced when the dependency is violated; and the Vermont evidence for the relationship.

The fifteen relationships are organized by upstream pillar — from Policy outward, then Technology, Economics, Clinical, and Equity. Operations receives dependencies but has no outbound dependencies in the primary framework: it is the execution layer that all other pillars feed into, not a source of enabling conditions for others. This is not because Operations is passive — it is because its function is to translate the outputs of the other five pillars into implemented programs, not to generate the conditions on which other pillars depend.

Three relationships are designated critical path: Policy → Technology, Policy → Economics, and Technology → Economics. These are the dependencies most consequential to the execution sequence, because their violation produces the most catastrophic and least reversible failure modes. They are discussed first within their upstream pillar sections and flagged in the master dependency table at the close of this chapter.

Relationship Type	Definition	Direction of Causality	Failure Mode When Violated
Enables (critical path)	The upstream pillar's presence is required for the downstream pillar to perform its core function. If the enabling pillar is absent or underdeveloped, the downstream pillar's core function is disabled — not merely weakened.	Upstream → downstream. The enabling pillar must be in place before the downstream pillar is activated.	Core function disabled. Interventions that look correct in design produce no structural result because the mandatory architecture or capital that would make them operational is absent.
Requires	The downstream pillar's management function depends on a specific output the upstream pillar must produce. The downstream pillar can exist without the upstream output but cannot be managed without it.	Upstream → downstream. The required output must be available before the downstream pillar's management functions are activated.	'Managing blind.' The intervention is real, the accountability is real, but the organization lacks the information to manage performance under it. Unexpected losses, undetected quality gaps, inability to dispute errors.
Constrains	The downstream pillar imposes design requirements on the upstream pillar that must be	Downstream → upstream. The constraining relationship	Technically functional, structurally unjust. Payment models systematically

	built in from the beginning, not retrofitted after deployment.	runs against the primary direction of authority. Equity constrains Economics and Clinical even though it comes after them in the execution sequence.	penalize safety-net providers. Clinical programs reach the well-served while leaving the underserved behind. Retrofitting equity after deployment is expensive and politically difficult.
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Figure 1A.1 — The three types of inter-pillar dependency relationship. Source: six-pillar framework documentation.

II. Policy Pillar — Outbound Dependencies (4 Relationships)

The Policy pillar has the highest outbound dependency count in the framework: four directed relationships, two of which are critical-path enablers. This reflects Policy's structural position as the pillar that creates the conditions under which every other pillar's mandatory functions can operate. Without the Policy pillar's mandatory architecture, the other five pillars are either aspirational programs or voluntary experiments — neither of which has produced structural healthcare transformation at scale.

Dependency 1: Policy → Technology (Enables — Critical Path)

Mechanism: The Policy pillar secures the transformation capital — federal grant awards, federal-state agreements, state appropriations — that funds the Technology build. Analytics platforms, health information exchange upgrades, population health management software, and clinical technology grants are capital-intensive investments that no individual hospital or health system can finance independently at the scale required for statewide transformation. The Policy pillar's federal relationship work — securing CMMI model agreements, rural health transformation grants, and HITECH-successor programs — is the funding mechanism for the Technology pillar.

Failure mode: Without Policy capital, the Technology build is either unfunded or fragmented into hospital-level investments that cannot achieve the statewide interoperability required for population health management. Individual hospitals invest in EHR systems; no one invests in the exchange layer that makes those systems collectively useful. This is the pre-HITECH national EHR story: millions of dollars in individual system investments producing a collection of data silos with no connectivity.

Vermont evidence: Vermont's \$195 million Rural Health Transformation Program award (December 2025) is the direct product of Act 167's statutory mandate, the AHEAD Model State Agreement, and years of federal relationship cultivation by the Agency of Human Services. The RHT award funds AI scribe technology grants, remote patient monitoring equipment, broadband infrastructure, and Vermont's CIN shared services — all Technology pillar investments that would not otherwise be financially feasible for Vermont's critical-access hospital system. The policy work preceded and enabled the technology investment.

Dependency 2: Policy → Economics (Enables — Critical Path)

Mechanism: Mandatory payment reform requires a statutory and regulatory architecture that converts voluntary payment model participation into binding obligation. Without this mandatory architecture, the highest-cost actors — the hospitals and health systems most responsible for total cost of care — rationally decline to participate in payment models that hold them accountable for costs they currently externalize. The Policy pillar's mandatory architecture is what transforms the Economics pillar from a voluntary experiment into structural payment reform.

Failure mode: Voluntary payment reform. The empirical record is unambiguous: voluntary ACO participation from 2010 to 2025 produced a national ACO ecosystem in which health systems with favorable patient populations and efficient existing operations participated, while the health systems most responsible for high costs declined. CMMI's fifty-plus voluntary models produced statistically significant savings in a small minority of cases. The models that showed durable results shared one feature: some mechanism of consequential participation.

Vermont evidence: OneCare Vermont operated as a voluntary ACO for nearly a decade. The University of Vermont Medical Center — Vermont's largest hospital, with approximately 50% of market share and the highest commercial rates in the state — declined to participate in OneCare's total cost of care accountability structure. The result was an ACO

accountability model that did not cover the actor most responsible for Vermont's cost crisis. Act 68 (2025) mandates global budgets covering all fourteen Vermont hospitals. The policy instrument — mandatory participation under GMCB authority — is what the decade of voluntary reform was missing.

Dependency 3: Policy → Clinical (Enables)

Mechanism: Clinical transformation at scale requires statutory authorization and program funding that exists only through policy action. Certificate of Need reform, scope-of-practice legislation, telehealth parity mandates, and CCBHC certification authority are all Policy pillar products that enable the Clinical pillar's programs. More fundamentally, the clinical redesign required for population health management requires that the policy environment mandate population health planning — otherwise clinical programs remain voluntary additions to a system that has no structural reason to invest in them at scale.

Failure mode: Clinical programs remain pilots. Without statutory authority requiring statewide clinical redesign — hospital transformation plans, population health commitments, CCBHC certification requirements — clinical innovation stays contained in institutions that choose to pursue it. Blueprint-style primary care transformation becomes available in willing practices but does not achieve population-level coverage.

Vermont evidence: Vermont's fourteen hospital transformation plans — 148 operational commitments submitted in January 2026 — exist because Act 167 requires them. The Blueprint for Health's expansion to nearly universal primary care coverage in Vermont was driven by GMCB requirements tied to budget approvals. The CCBHC expansion is driven by federal certification authority accessed through state policy action. Clinical transformation at this scale does not happen voluntarily.

Dependency 4: Policy → Operations (Enables)

Mechanism: The Policy pillar's statutory mandates define what the Operations pillar must execute. Without specific statutory timelines, accountability structures, and reporting requirements, the Operations pillar lacks the urgency and specificity required for serious transformation implementation. Acts 167 and 68 are, in operational terms, a detailed implementation mandate: they specify what must be done, by when, by whom, with what reporting requirements, and with what consequences for non-compliance. The Operations pillar's PMO functions are organized around executing those mandates.

Failure mode: The document graveyard. Without statutory mandate, transformation plans are produced on advisory timelines with advisory accountability. The statewide strategic plan that Act 167 requires Vermont to deliver to the legislature by December 2028 has a deadline that cannot be deferred without legislative consequence. Transformation planning without statutory deadline and accountability has a consistent historical outcome: plans that are carefully produced, widely circulated, and never implemented.

Vermont evidence: The AHS monthly transformation reports — required under Act 167 — are a direct product of the statutory reporting mandate. Without that mandate, AHS transformation activities would be reported at the agency's discretion, on a timeline the agency controls, with metrics the agency selects. The mandate converts voluntary reporting into a public accountability structure that creates organizational pressure for implementation.

III. Technology Pillar — Outbound Dependencies (3 Relationships)

The Technology pillar has three outbound dependencies, all in the 'requires' category: Economics, Clinical, and Equity each require specific Technology pillar outputs to perform their management functions. Technology has no constraining outbound dependencies because it does not impose design requirements on other pillars — it provides the information substrate that makes other pillars' management functions possible. The Technology → Economics dependency is the most consequential in the entire framework, because its violation produced the most documented and costly failure in Vermont's transformation history.

Dependency 5: Technology → Economics (Requires — Critical Path)

Mechanism: Value-based payment models — global budgets, total cost of care contracts, reference-based pricing enforcement — require real-time or near-real-time data on patient attribution, utilization patterns, cost trajectories, quality metrics, and care gaps to be managed operationally. Without this data substrate, the payment model exists as a financial

accountability structure with no operational management capability. Organizations sign agreements that commit them to cost and quality performance but cannot see their performance in time to intervene.

“Managing a global budget without integrated clinical and claims data is not bold reform. It is financial accountability without management capability — the conditions under which organizations can only respond to failures that have already occurred, not prevent them.” — HTR Framework Analysis

Failure mode: The 'managing blind' failure mode. OneCare Vermont operated with 60-to-90-day-old claims data as its primary performance signal. By the time care managers received attribution data identifying patients as high-risk, those patients had frequently already been hospitalized. The cost had been incurred. The intervention opportunity had passed. Care managers were responding to a historical record, not managing a current population. The global budget accountability was real; the management capability to perform under it was structurally absent.

Vermont evidence: The AHS-GMCB analytics vendor procurement (McKinsey, early 2026) represents Vermont's belated recognition that the AHEAD Model's global budgets — launching January 2027 for Medicare — cannot be managed without integrated analytics. The timing creates Vermont's most significant current sequencing risk: the analytics platform must be operational before the global budget goes live, not after. Vermont's VHCURES all-payer claims database provides the foundational data but arrives with a two-year lag in standard reporting — a lag that makes it inadequate as the sole operational management tool for a live global budget.

Dependency 6: Technology → Clinical (Requires)

Mechanism: Clinical population health management — risk stratification, care gap identification, care management targeting, quality measurement, and outcome tracking — requires data infrastructure that makes population-level visibility possible. Without the Technology pillar's analytics and HIE capabilities, clinical programs serve the patients who present for care rather than the patients the population health model identifies as highest-risk and highest-priority. The clinical program is real; its population health function is disabled.

Failure mode: Clinical programs that work for engaged patients and miss the population. Care management programs without risk stratification data identify patients for intensive management through provider referral — capturing patients already in the care system — rather than through predictive modeling that identifies patients approaching crisis before they present to an emergency department. The clinical investment is real; the population health return is a fraction of what integrated data would enable.

Vermont evidence: Vermont's Blueprint for Health community health teams operate with panel data from the Blueprint's registry system — a relatively sophisticated data tool by national standards. The Blueprint's documented 5.8:1 ROI reflects the advantage of data-enabled care management compared to referral-based programs. The next generation of clinical program — AHEAD-funded primary care, Vermont CIN shared care management — requires the analytics vendor's population health platform to achieve a comparable data advantage at system scale rather than practice level.

Dependency 7: Technology → Equity (Requires)

Mechanism: Equity measurement — identifying whether transformation benefits are reaching all populations — requires data stratified by race, ethnicity, income, geography, language, and disability status. Without this stratification capability in the Technology pillar's data infrastructure, equity analysis is limited to aggregate outcomes that conceal disparities. A transformation that improves average outcomes while widening disparities cannot be detected, and therefore cannot be corrected, without stratified data.

Failure mode: Average outcome improvement masking disparity widening. Health equity goals are embedded in statutory language but cannot be measured with the available data. Reporting shows improving statewide averages. Rural communities, communities of color, and low-income populations experience unchanged or worsening conditions that are invisible in aggregate reporting.

Vermont evidence: Vermont's VHCURES all-payer claims database has documented gaps in race and ethnicity data capture that limit the demographic stratification required for equity accountability under Acts 167 and 68. The Vermont Department of Health Health Equity Data Report (January 2025) identified these gaps as a priority for the analytics vendor procurement. Vermont's fourteen HSA-level geographic equity monitoring — a requirement of Act 68 — depends on the analytics platform's ability to produce HSA-stratified outcome data, which VHCURES in its current form cannot produce in real time.

IV. Economics Pillar — Outbound Dependencies (3 Relationships)

The Economics pillar's outbound dependencies run to Clinical, Equity, and Operations. All three are 'requires' relationships: the payment architecture must be in place before clinical programs can achieve their full population health function, before equity measurement can assess whether financial incentives are producing equitable results, and before operational redesign can be calibrated to the payment environment it will actually work within.

Dependency 8: Economics → Clinical (Requires)

Mechanism: Clinical redesign for population health management — prevention-focused primary care, behavioral health integration, care management at scale — is financially rational only when the payment model rewards these activities. Under fee-for-service, the primary care visit that prevents a hospitalization generates no revenue from the prevention. The hospital that reduces its avoidable admission rate reduces its own income. Clinical programs that produce population health improvement must operate against financial incentives that punish exactly the outcomes they are designed to produce. Clinical transformation requires the Economics pillar to have already converted these perverse incentives into aligned ones.

Failure mode: Clinical redesign that reverts. Clinical programs produce outcomes but cannot capture their financial return under the existing payment model. Hospitals and practices that invest in prevention and care coordination find that the investment produces population health improvement — and revenue reduction. Programs are discontinued when grant funding expires. Evidence-based clinical models demonstrate effectiveness in funded pilots and fail to scale because the payment system makes scaling financially irrational.

Vermont evidence: Vermont's Blueprint for Health has sustained itself because Blueprint per-member per-month payments create a partial financial return on prevention investment. But Blueprint payments cover only a fraction of the clinical investment required for comprehensive population health management. The AHEAD Model's primary care investment — replacing Blueprint's Medicare payments with AHEAD prospective payments — is the Economics pillar instrument designed to complete the financial alignment that Blueprint partially achieved. Until AHEAD's payment model is live, Clinical pillar investment in AHEAD-scale primary care transformation operates on partial incentive alignment.

Dependency 9: Economics → Equity (Requires)

Mechanism: The Economics pillar's payment design determines whether financial incentives systematically disadvantage safety-net providers — hospitals and practices serving high proportions of Medicaid, dual-eligible, and uninsured patients in communities with high social risk. Global budgets set without social risk adjustment hold safety-net providers accountable for costs driven by their population's social conditions rather than their operational efficiency. The Equity pillar's Stage 5 assessment of whether financial incentives produced equitable results requires the payment model to already be live and producing measurable financial signals.

Failure mode: Payment design that penalizes serving disadvantaged populations. Safety-net providers face budget targets that do not reflect their population's underlying risk. They appear to be inefficient — exceeding benchmark costs — when they are in fact serving populations whose social conditions generate systematically higher healthcare utilization. Financial pressure from a poorly designed global budget produces access restriction in exactly the communities the equity framework was designed to protect.

Vermont evidence: The Northeast Kingdom — Vermont's most economically disadvantaged region, with an uninsurance rate of 6-8% against a 3% statewide average — is served by critical-access hospitals whose cost structures reflect their population's high social risk. Act 68's global budget methodology must account for this population risk differential, or the budget will systematically penalize NE Kingdom hospitals for their geography. Vermont's AHEAD State Agreement includes social risk adjustment as a requirement; the GMCB's implementation methodology for that adjustment is the critical design choice.

Dependency 10: Economics → Operations (Requires)

Mechanism: Operational redesign — administrative simplification, shared services, revenue cycle transformation, workforce redeployment — must be calibrated to the payment environment it is executing within. The administrative operations optimized for fee-for-service billing — prior authorization management, claims coding, utilization review — are different from the administrative operations optimized for global budget management, which require population health analytics, care management coordination, and TCOC reporting. Building the latter before the former is retired wastes capital and creates organizational confusion.

Failure mode: Operations optimized for the wrong payment model. Hospitals that invest in administrative transformation before their payment model is defined build infrastructure for a system that may be substantially redesigned before the infrastructure is fully deployed. Revenue cycle staff retrained for value-based contract management in a system that remains predominantly fee-for-service have been trained for work that does not yet exist at scale.

Vermont evidence: Vermont's critical-access hospitals spend \$2,730 per adjusted discharge on administration against a \$1,427 national benchmark — a \$1,303 per-discharge gap representing the single largest direct savings opportunity in Vermont's transformation architecture. But capturing that gap requires knowing what the hospitals are administering: fee-for-service billing operations, global budget management operations, or a hybrid of both during the transition period. The RHRC's administrative simplification work — shared billing services, consolidated prior authorization, CON streamlining — must be designed around the payment model that will be live when the shared services are operational.

V. Clinical Pillar — Outbound Dependencies (2 Relationships)

The Clinical pillar generates two outbound dependencies, both to the Equity pillar. This reflects Clinical's position as the mechanism through which population health commitments are either fulfilled or failed: clinical programs that are effective on average but inequitable in their reach produce the most insidious form of transformation failure — measurable population health improvement at the aggregate level that conceals the persistence or widening of disparities.

Dependency 11: Clinical → Equity (Requires)

Mechanism: Equity measurement at Stage 5 requires that clinical programs have been deployed at sufficient scale and duration to produce measurable population health signals stratified by race, income, geography, and other equity dimensions. Equity analysis of clinical program reach — which communities are receiving Blueprint-equivalent primary care, which communities are accessing CCBHC services, which populations are benefiting from care management — requires that the Clinical pillar's programs be operational. The Equity pillar cannot assess what has not yet happened.

Failure mode: Equity measurement of programs too nascent to produce measurable signals. Equity reporting shows insufficient data to assess disparities. Equity goals remain aspirational because the clinical programs they depend on are still in early deployment. The equity framework is structurally sound but analytically empty.

Vermont evidence: Vermont's CCBHC expansion — seven certified entities by mid-2026, targeting geographic coverage of behavioral health access — is the Clinical pillar investment whose equity implications the Equity pillar's Stage 5 assessment will measure. Are CCBHCs located in communities with the highest unmet behavioral health need? Are they serving Medicaid and uninsured patients at rates proportionate to community need? These questions require the CCBHC programs to be operational and the analytics platform to be producing stratified utilization data.

Dependency 12: Clinical → Operations (Requires)

Mechanism: Clinical program implementation at scale — Blueprint community health teams in every primary care practice, CCBHC services in every underserved community, care management for every high-risk patient — requires operational infrastructure: trained workforce, referral pathways, documentation systems, quality measurement processes, and supervisory structures. The Operations pillar's execution function cannot be fully designed until the Clinical pillar has specified what programs must be executed. Clinical program design precedes operational build-out, not the reverse.

Failure mode: Operational infrastructure built for programs that are subsequently redesigned, or clinical programs specified without operational feasibility assessment. Vermont's fourteen hospital transformation plans commit to 148 specific clinical initiatives. Each of those initiatives requires an operational plan: who does it, with what training, using what data system, measured against what metric, with what consequence if not implemented. The operational plans must follow the clinical commitments — not precede them.

Vermont evidence: The RHRC's technical assistance methodology — the Rural Health Redesign Center's engagement with Vermont's hospitals — is organized around translating clinical transformation commitments into operational implementation plans. The RHRC does not specify what clinical programs hospitals should pursue; it helps hospitals operationalize the programs they have committed to. This is the correct sequencing of the Clinical → Operations relationship: clinical program definition first, operational build-out second.

VI. Equity Pillar — Outbound Dependencies (3 Relationships)

The Equity pillar's three outbound dependencies are all 'constrains' relationships — the only constraining dependencies in the framework. This is the defining structural feature of how equity works in the portfolio: equity does not enable or require other pillars' functions, it constrains their design. Payment architecture, clinical program deployment, and operational prioritization must all be designed from the outset with equity requirements built in. Retrofitting equity after these systems are live is expensive, technically difficult, and politically contentious.

Dependency 13: Equity → Economics (Constrains)

Mechanism: The Equity pillar imposes social risk adjustment requirements on the Economics pillar's global budget methodology. Without social risk adjustment, global budgets systematically disadvantage hospitals serving populations with high social determinants of health burden — high poverty rates, high housing instability, high food insecurity — whose healthcare utilization is driven by social conditions the hospital cannot control. The constraint must be embedded in the payment design process, before the budget methodology is finalized, not after hospitals begin experiencing its inequitable effects.

Failure mode: Global budgets that redistribute financial pressure toward safety-net providers. Hospitals serving disadvantaged populations are penalized for their population's social risk, not their operational inefficiency. This creates the most politically damaging version of transformation failure: a reform designed to improve healthcare produces financial distress in the hospitals serving the most vulnerable communities.

Vermont evidence: Vermont's AHEAD State Agreement requires social risk adjustment in the AHEAD global budget methodology. The GMCB's FY2026 budget cycle began developing a social risk adjustment framework for the transition to mandatory global budgets. The specific methodology — what social risk variables are included, how they are weighted, how frequently they are updated — is the most consequential technical design choice in Vermont's current transformation architecture for the equity implications of the payment model.

Dependency 14: Equity → Clinical (Constrains)

Mechanism: The Equity pillar imposes geographic and demographic reach requirements on Clinical pillar program deployment. Clinical programs must be designed and sited to reach underserved communities, not merely available in communities that already have adequate healthcare infrastructure. This means CCBHCs must be certified in underserved areas, not only in communities with existing behavioral health infrastructure. Blueprint-equivalent primary care must reach rural communities with workforce shortages, not only well-served urban areas. Cultural competency requirements must be embedded in clinical program design, not treated as optional enhancements.

Failure mode: Clinical transformation that serves the well-served. New clinical programs are deployed in communities with existing healthcare infrastructure, willing providers, and engaged patient populations — producing measurable outcomes in the communities that needed improvement least. Rural communities, communities of color, and low-income populations — the communities with the greatest gap between current conditions and transformation goals — receive less investment because the equity constraint was not built into the clinical deployment strategy.

Vermont evidence: Vermont's Northeast Kingdom — Caledonia, Essex, and Orleans counties — faces the state's most severe combination of healthcare access deficit and social risk. The NE Kingdom's uninsurance rate of 6-8%, its shortage of primary care providers, and its geographic isolation create a clinical investment priority that equity constraints require to be embedded in CCBHC siting decisions, AHEAD primary care investment targeting, and Rural Emergency Hospital model deployment. Act 68's geographic equity monitoring requirement at the HSA level is the policy mechanism that makes this constraint operational.

Dependency 15: Equity → Operations (Constrains)

Mechanism: The Equity pillar imposes prioritization requirements on the Operations pillar's implementation sequencing. When organizational capacity is limited — when AHS cannot simultaneously build every program at full speed — equity constraints require that underserved communities receive prioritized attention, not deferred attention. The HSA Coordinator model, which assigns operational capacity to specific Hospital Service Areas, must be deployed first in the HSAs with the greatest equity deficits. Operational prioritization driven by administrative convenience rather than equity need produces a transformation that reaches communities in order of their existing organizational capacity — which is the inverse of equity priority.

Failure mode: Operational sequencing that follows organizational convenience. Technical assistance is provided first to hospitals with existing transformation capacity — those with dedicated quality improvement staff, experienced leadership, and organizational cultures receptive to change. Critical-access hospitals in underserved communities — those with the least existing capacity and the greatest transformation need — receive assistance last, when organizational bandwidth permits. The equity gap widens during the transformation period before it narrows, if it narrows at all.

Vermont evidence: Vermont's RHRC engagement prioritization — which hospitals receive intensive technical assistance first — is an operational decision with direct equity implications. The RHRC's mandate under the RHT Program explicitly prioritizes rural and critical-access hospitals. Vermont's proposed HSA Coordinator positions — AHS staff assigned to provide operational support at the hospital service area level — are designed to operationalize the equity constraint on operations by embedding state capacity in the communities with the greatest organizational need.

VII. The Master Dependency Table

The following table consolidates all fifteen directed dependency relationships, their types, mechanisms, and failure modes. It is designed as the analytical reference for the execution sequence developed in Chapter 2 and for the risk assessment that any reform architect or health system leader must conduct before making investment and sequencing decisions.

From Pillar	To Pillar	Type	Mechanism	Failure Mode	Vermont Evidence
Policy	Technology	Enables (Critical Path)	Policy secures the transformation capital — federal grants and federal-state agreements — that funds the statewide Technology build at a scale no individual institution can finance independently.	Fragmented, individually funded technology investments producing disconnected data silos. No statewide interoperability. Population health management remains analytically impossible.	\$195M RHT award funds AI scribe, RPM, broadband, and CIN — all enabled by the AHEAD State Agreement and Act 167 policy work.
Policy	Economics	Enables (Critical Path)	Mandatory authority converts voluntary payment model participation into binding obligation, preventing the highest-cost actors from opting out of accountability.	Voluntary ACO ecosystem. High-cost actors decline. Results limited to willing participants with favorable populations. No structural cost reduction.	OneCare: voluntary decade, UVMHC not participating. Act 68: mandatory global budgets, all 14 hospitals covered under GMCB authority.
Policy	Clinical	Enables	Statutory authorization, program funding, and statewide planning requirements mandate clinical transformation at population scale rather than individual institution choice.	Clinical innovation stays in willing institutions. No population-level coverage. Programs remain pilots indefinitely.	148 hospital plan commitments exist because Act 167 requires them. Blueprint near-universal coverage driven by GMCB budget conditions.
Policy	Operations	Enables	Statutory timelines and accountability structures create the urgency and	Document graveyard. Plans produced on advisory timelines with	Act 167 monthly reporting mandate

			specificity required for serious transformation implementation at an organizational level.	no binding accountability. Widely circulated, never implemented.	converts voluntary progress tracking into public accountability structure.
Technology	Economics	Requires (Critical Path)	VBC contracts and global budgets require real-time attribution, utilization, TCOC, and quality data to be managed operationally. Without this substrate, financial accountability exists without management capability.	'Managing blind.' Care managers working from 60-90 day old data. Interventions arrive after costs are incurred. Performance management is retrospective, not prospective.	OneCare's canonical failure. AHS-GMCB analytics vendor procurement required before AHEAD January 2027 global budget launch.
Technology	Clinical	Requires	Clinical population health management requires risk stratification, care gap identification, and care management targeting data that only integrated data infrastructure can produce at population scale.	Clinical programs serve presenting patients, not population priorities. High-risk patients identified after crisis presentation rather than before. Population health return is a fraction of data-enabled potential.	Blueprint registry data enables 5.8:1 ROI. AHEAD-scale clinical programs require analytics vendor's population health platform for comparable advantage at system level.
Technology	Equity	Requires	Equity measurement requires data stratified by race, ethnicity, income, geography, and disability. Without stratification capability, equity analysis is limited to aggregates that conceal disparities.	Average outcome improvement masking disparity widening. Equity goals embedded in statute but unmeasurable. Improving statewide averages conceal unchanged or worsening conditions in disadvantaged communities.	VHCURES race/ethnicity data gaps limit demographic stratification. HSA-level geographic monitoring under Act 68 requires analytics platform VHCURES cannot currently produce in real time.
Economics	Clinical	Requires	Clinical redesign for population health is financially rational only when the payment model rewards prevention and care coordination. Under FFS, these activities reduce revenue.	Clinical redesign that reverts. Programs prove effective in funded pilots. Financial return cannot be captured under existing payment model. Programs discontinued when grant funding expires.	Blueprint PMPM creates partial alignment. AHEAD primary care payment completes it. Until AHEAD is live, clinical investment operates on incomplete

					incentive alignment.
Economics	Equity	Requires	Equity pillar's Stage 5 assessment of financial incentive equity requires the payment model to be live and producing measurable financial signals for assessment.	Equity measurement of a payment model not yet deployed. Equity goals aspirational because the economic intervention they assess has not yet occurred.	AHEAD global budgets must be live before equity assessment of their distributional effects is possible. Social risk adjustment audit requires post-launch data.
Economics	Operations	Requires	Operational redesign must be calibrated to the payment environment it will execute within. FFS operations and global budget operations are structurally different functions.	Operations optimized for the wrong payment model. Shared services and revenue cycle transformation built for a payment architecture that is subsequently redesigned.	\$1,303/discharge administrative gap. RHRC shared services design must target the global budget operational model, not a hybrid FFS-to-global budget transitional model.
Clinical	Equity	Requires	Equity measurement of clinical program reach requires programs to be operational at sufficient scale and duration to produce measurable stratified population health signals.	Equity measurement of programs too nascent to produce measurable signals. Equity framework structurally sound but analytically empty.	CCBHC geographic reach equity assessment requires seven-entity deployment to be operational and analytics platform producing stratified utilization data.
Clinical	Operations	Requires	Operational infrastructure for clinical programs must follow program specification. Clinical design defines what operations must execute; operational planning follows clinical commitment.	Operational infrastructure built for programs subsequently redesigned, or programs specified without operational feasibility assessment.	RHRC translates 148 clinical hospital plan commitments into implementation plans. Clinical commitment precedes operational design — the correct sequencing.
Equity	Economics	Constraints	Equity imposes social risk adjustment	Global budgets penalizing safety-net	AHEAD State Agreement

			requirements on global budget methodology from design outset. Must be embedded before methodology is finalized.	providers for serving high-SDOH populations. Financial pressure on exactly the hospitals serving the most vulnerable communities.	requires social risk adjustment. GNCB methodology development is the critical design choice for NE Kingdom hospital financial viability.
Equity	Clinical	Constraints	Equity imposes geographic and demographic reach requirements on clinical program siting and design from the outset, not as an afterthought.	Clinical transformation serving the well-served. Programs deployed in communities with existing infrastructure. Underserved communities receive less investment.	NE Kingdom priority for CCBHC siting, AHEAD primary care targeting, and REH model deployment must be embedded in clinical program design, not assessed after deployment.
Equity	Operations	Constraints	Equity imposes prioritization requirements on operational sequencing. Underserved communities must receive prioritized technical assistance and organizational support, not deferred attention.	Operational sequencing following organizational convenience. Hospitals with existing capacity receive support first. Equity gap widens during transformation before (if ever) narrowing.	RHRC mandate prioritizes rural and CAH hospitals. HSA Coordinator positions embed state capacity in highest-need communities. Operationalizes equity constraint on implementation sequencing.

Figure 1A.2 — All fifteen directed dependency relationships. Critical-path dependencies: Policy → Technology, Policy → Economics, Technology → Economics. Source: six-pillar framework documentation; Vermont experience 2010-2026; HTR Analysis (2026).

VIII. Failure Cascade Analysis — What Breaks When Any Pillar Is Absent

The dependency architecture generates specific, testable predictions about what fails when any pillar is absent, underfunded, or activated in the wrong sequence. The following analysis traces the cascade for each pillar, identifying the first-order and second-order failures produced by each gap.

Pillar Absent or Severely Underdeveloped	First-Order Failure	Second-Order Cascade
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Policy	Voluntary reform. High-cost actors opt out. No mandatory architecture for accountability.	Without mandatory Economics, technology investments serve willing participants only. Clinical programs remain pilots. Operations lacks statutory urgency. OneCare is the first-order and second-order failure simultaneously.
Technology	'Managing blind.' Global budgets and VBC contracts exist without operational management capability.	Economics produces unexpected financial losses (providers cannot manage what they cannot see). Clinical programs cannot risk-stratify or identify care gaps. Equity measurement is analytically impossible. Operations cannot measure implementation effectiveness.
Economics	Clinical redesign without aligned incentives. Prevention investment is financially penalized.	Clinical programs prove effective in pilots but cannot achieve financial sustainability or system-wide scale. Workforce is retrained for population health functions that the payment system continues to penalize. Operations builds infrastructure for programs that cannot sustain themselves.
Clinical	Financial pressure without clinical pathway. Global budgets create incentive to reduce costs with no mechanism to do so through improved population health.	Economics produces access restriction rather than population health improvement. Hospitals reduce high-cost care without substituting effective lower-cost alternatives. Equity worsens as restricted access falls hardest on disadvantaged populations.
Equity	Transformation that improves averages while widening disparities. Success metrics show improvement while disparate harm persists.	Payment models penalize safety-net providers (Economics without Equity constraint). Clinical programs concentrate in well-served communities (Clinical without Equity constraint). Political legitimacy of transformation collapses when disparity widening becomes visible.
Operations	The document graveyard. Technically correct reforms that cannot be implemented at scale. Strategy without execution.	Policy mandates produce legal exposure without operational compliance. Economics produces payment models that providers cannot operationally manage. Clinical programs are specified but not implemented. Technology platforms are procured and unused.

Figure 1A.3 — Failure cascade analysis by absent pillar. Source: HTR Analysis (2026); CMMI model evaluation literature; Vermont experience.

Key Concepts Introduced in This Chapter

Term	Definition
Dependency architecture	The complete map of fifteen directed inter-pillar relationships — enables, requires, and constrains — that determine how the six pillars interact and why their sequence is not optional.

Critical-path dependency	A dependency whose violation produces the most catastrophic and least reversible failure modes in the transformation system. The three critical-path dependencies are Policy → Technology, Policy → Economics, and Technology → Economics.
'Managing blind' failure mode	The result of activating Economics-as-management (global budgets, VBC contracts) before Technology is operational: financial accountability without management capability. Canonically demonstrated by OneCare Vermont's operation on 60-to-90-day-old claims data.
First-order failure	The direct consequence of a pillar's absence: the specific function that fails when a pillar is not in place. Policy absent → voluntary reform. Technology absent → managing blind.
Second-order cascade	The downstream failures produced by a first-order failure: the additional pillar functions disabled when the first-order failure propagates through the dependency architecture.
Social risk adjustment	The modification of global budget targets or VBC benchmarks to account for the social risk characteristics of a provider's patient population, preventing payment models from penalizing safety-net providers for serving high-SDOH communities. A product of the Equity → Economics constraining dependency.
Constraining dependency	A dependency relationship in which the downstream pillar imposes design requirements on the upstream pillar from the outset. Equity's three constraining dependencies (→ Economics, → Clinical, → Operations) are the mechanism through which equity becomes a design requirement rather than a measurement afterthought.
Inbound dependency count	The number of other pillars that depend on a given pillar's outputs. High inbound dependency count indicates high leverage: delays or failures in a high-inbound-count pillar simultaneously impair multiple other pillar functions. Technology has the highest inbound dependency count in the framework (Economics, Clinical, and Equity all require Technology outputs).

Key Concepts — Chapter 1A. Full glossary available in the Index.

Sources

six-pillar framework documentation; Act 167 of 2022 (Vermont); Act 68 of 2025 (Vermont); Oliver Wyman Healthcare and Life Sciences Group, Act 167 Community Engagement: Recommendations (August 2024, revised October 2024, January 2025); Vermont Agency of Human Services Health Care System Transformation Reports (August 2025, November 2025); Vermont Rural Health Transformation Program Application (November 2025); Vermont AHEAD State Agreement, Cohort 2 (January 17, 2025); GMCB Hospital Budget Orders and Rate Review Records (FY2023-FY2026); OneCare Vermont Annual Reports (2017-2024); CMMI model evaluation, Congressional Budget Office (2023); Vermont Blueprint for Health Annual Report and ROI Study; Vermont Department of Health Health Equity Data Report (January 2025); AHS Hospital Transformation Plans Analysis (January 2026); Vermont VHCURES All-Payer Claims Database documentation; VTTL/VHIE utilization reports; HTR Framework Analysis (2026).